

Project Title:

Identifying High-Potential but Under-Engaged Customers Using Behavioral and Demographic Data

This project aims to analyze customer data to identify high-potential customers who are currently under-engaged. The main research question is whether customer segmentation based on behavioral and demographic characteristics can reveal such customers, and how these insights can be used to improve targeting strategies and marketing effectiveness.

The dataset used in this project will be the Customer Personality Analysis Dataset, obtained from Kaggle. The dataset contains detailed customer-level information, including demographic variables such as age (derived from year of birth), income, education, and marital status, as well as behavioral variables such as purchase frequency, recency, spending across product categories, and campaign responses.

The dataset includes approximately 2,000 customers with multiple numerical and categorical attributes, making it suitable for both clustering and predictive modeling. Since the dataset is publicly available, it will be enriched through feature engineering by constructing additional variables such as total spending, RFM (Recency, Frequency, Monetary) metrics, and normalized behavioral indicators to better capture customer engagement and potential.

The analysis will begin with data preprocessing and exploratory data analysis (EDA) to understand patterns in customer behavior. Customer segmentation will then be performed using clustering methods such as K-Means and hierarchical clustering. The optimal number of clusters will be determined using the Elbow Method and Silhouette Score. The resulting segments will be analyzed to identify high-potential but under-engaged customers, characterized by strong purchasing capacity but relatively low recent activity or engagement.

In addition, a classification model will be developed to predict whether a customer belongs to this high-potential segment, enabling forward-looking targeting decisions. The findings will be translated into actionable strategies, such as personalized campaigns or re-engagement efforts for specific customer groups.

The expected outcome of this project is to uncover actionable customer segments and provide data-driven recommendations that can improve marketing efficiency, customer engagement, and resource allocation.